

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 1            |        |        |        |        |        |        |
| DLT rate              | 0.3    | 0.35   | 0.4    | 0.45   | 0.5    |        |
| Efficacy rate         | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |        |
| Utility 2             | -0.2   | -0.3   | -0.3   | -0.4   | -0.4   |        |
| Utility 3             | 31     | 29.6   | 28.2   | 26.8   | 25.4   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 12.4   | 4.9    | 2.6    | 1.1    | 0.4    | 78.6   |
| U-BOIN                | 10.6   | 15.1   | 18.2   | 19.5   | 17.2   | 19.4   |
| EffTox                | 0      | 0      | 4      | 1      | 2      | 92     |
| AIDE one-stage (U2)   | 8.6    | 3      | 2.4    | 0.5    | 0.8    | 84.7   |
| AIDE one-stage (U3)   | 7.9    | 3.6    | 3.1    | 0.6    | 0.7    | 84.1   |
| AIDE two-stage (U2)   | 5.9    | 3.5    | 0.6    | 0.6    | 0.2    | 89.2   |
| AIDE two-stage (U3)   | 6.1    | 3.7    | 0.7    | 0.2    | 0.2    | 89.1   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 7.5    | 5      | 3.3    | 2      | 1      |        |
| U-BOIN                | 9      | 5.3    | 4.7    | 4.3    | 3.9    |        |
| EffTox                | 4.4    | 3.2    | 2.7    | 2      | 1.8    |        |
| AIDE one-stage (U2)   | 7.5    | 4.8    | 2.2    | 1.3    | 0.4    |        |
| AIDE one-stage (U3)   | 7.6    | 4.8    | 2.2    | 1.3    | 0.4    |        |
| AIDE two-stage (U2)   | 7.8    | 3      | 1.9    | 0.9    | 0.4    |        |
| AIDE two-stage (U3)   | 7.8    | 3      | 2      | 0.9    | 0.4    |        |
| Scenario 2            |        |        |        |        |        |        |
| DLT rate              | 0.15   | 0.2    | 0.3    | 0.35   | 0.45   |        |
| Efficacy rate         | 0.2    | 0.25   | 0.5    | 0.5    | 0.5    |        |
| Utility 2             | 0.1    | 0.1    | 0.2    | 0.1    | 0.1    |        |
| Utility 3             | 46     | 47     | 58     | 56     | 52     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 29.4   | 20.9   | 31.3   | 8.9    | 2.2    | 7.3    |
| U-BOIN                | 11.7   | 10.5   | 41.1   | 22.9   | 12.2   | 1.6    |
| EffTox                | 35     | 12     | 15     | 14     | 16     | 8      |
| AIDE one-stage (U2)   | 20     | 16.3   | 37.7   | 14.6   | 2.2    | 9.2    |
| AIDE one-stage (U3)   | 24.7   | 17.3   | 34.6   | 12.1   | 1.9    | 9.4    |
| AIDE two-stage (U2)   | 25     | 14.4   | 32.6   | 11.2   | 3.7    | 13.1   |
| AIDE two-stage (U3)   | 28     | 15.4   | 31     | 10     | 2.6    | 13     |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 8.1    | 7.5    | 8.4    | 3.7    | 1.2    |        |
| U-BOIN                | 8.5    | 7.1    | 8.3    | 4.1    | 1.7    |        |
| EffTox                | 27.5   | 7.2    | 7.1    | 6.2    | 8.7    |        |
| AIDE one-stage (U2)   | 8.2    | 6.7    | 8.7    | 3.9    | 0.9    |        |
| AIDE one-stage (U3)   | 8.7    | 6.8    | 8.6    | 3.6    | 0.7    |        |
| AIDE two-stage (U2)   | 9.5    | 5      | 7.2    | 3.9    | 2.1    |        |
| AIDE two-stage (U3)   | 9.6    | 4.9    | 7.3    | 3.8    | 2.2    |        |

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda\_T = 0.3; U3: Type = 3, Lambda\_T = 1.

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 3            |        |        |        |        |        |        |
| DLT rate              | 0.15   | 0.2    | 0.3    | 0.35   | 0.45   |        |
| Efficacy rate         | 0.1    | 0.4    | 0.4    | 0.4    | 0.4    |        |
| Utility 2             | -0.1   | 0.2    | 0.1    | 0.1    | -0.1   |        |
| Utility 3             | 40     | 56     | 52     | 50     | 46     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 7.1    | 56.8   | 18.2   | 6.2    | 1.3    | 10.4   |
| U-BOIN                | 2.7    | 43.6   | 24.6   | 18.5   | 9      | 1.6    |
| EffTox                | 18     | 33     | 14     | 13     | 14     | 8      |
| AIDE one-stage (U2)   | 6.3    | 36.8   | 24.7   | 13.1   | 2.3    | 16.8   |
| AIDE one-stage (U3)   | 7.7    | 42.4   | 22.3   | 9      | 1.7    | 16.9   |
| AIDE two-stage (U2)   | 7.1    | 36.4   | 21.2   | 8.2    | 2.6    | 24.5   |
| AIDE two-stage (U3)   | 8.7    | 40.3   | 18.5   | 6.5    | 1.6    | 24.4   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 5.5    | 11.8   | 5.9    | 3.4    | 1.4    |        |
| U-BOIN                | 7      | 10.9   | 6.7    | 3.5    | 1.7    |        |
| EffTox                | 17.3   | 15.5   | 9.1    | 5.9    | 8.2    |        |
| AIDE one-stage (U2)   | 5.4    | 10.1   | 6.5    | 3.8    | 1.1    |        |
| AIDE one-stage (U3)   | 5.4    | 10.9   | 6.4    | 3.4    | 0.9    |        |
| AIDE two-stage (U2)   | 6.3    | 7.4    | 6.1    | 3.6    | 2      |        |
| AIDE two-stage (U3)   | 6.3    | 7.3    | 6.2    | 3.5    | 2      |        |
| Scenario 4            |        |        |        |        |        |        |
| DLT rate              | 0.05   | 0.1    | 0.18   | 0.3    | 0.4    |        |
| Efficacy rate         | 0.05   | 0.1    | 0.25   | 0.5    | 0.55   |        |
| Utility 2             | 0      | 0      | 0.1    | 0.2    | 0.1    |        |
| Utility 3             | 41     | 42     | 47.8   | 58     | 57     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 10.9   | 11.9   | 25.6   | 36.7   | 9      | 5.9    |
| U-BOIN                | 2.2    | 5.6    | 17.7   | 45.6   | 28.8   | 0.1    |
| EffTox                | 0      | 2      | 25     | 33     | 33     | 8      |
| AIDE one-stage (U2)   | 5.7    | 4.1    | 16.3   | 52.8   | 16.9   | 4.2    |
| AIDE one-stage (U3)   | 8.2    | 6.4    | 16.1   | 51.3   | 13.4   | 4.6    |
| AIDE two-stage (U2)   | 4.5    | 5.9    | 18.3   | 43.4   | 20.4   | 7.5    |
| AIDE two-stage (U3)   | 7.9    | 7.4    | 18     | 42.4   | 16.8   | 7.5    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 4.3    | 5.5    | 7.3    | 8.9    | 3.5    |        |
| U-BOIN                | 5      | 6.3    | 7.4    | 7.6    | 4      |        |
| EffTox                | 4.6    | 4.7    | 14.1   | 12.9   | 19.9   |        |
| AIDE one-stage (U2)   | 3.8    | 4.3    | 6.5    | 10.6   | 3.9    |        |
| AIDE one-stage (U3)   | 3.9    | 4.4    | 6.6    | 10.6   | 3.5    |        |
| AIDE two-stage (U2)   | 4      | 4      | 6.3    | 8      | 6.5    |        |
| AIDE two-stage (U3)   | 4      | 4      | 6.3    | 8.1    | 6.5    |        |

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda\_T = 0.3; U3: Type = 3, Lambda\_T = 1.

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 5            |        |        |        |        |        |        |
| DLT rate              | 0.07   | 0.12   | 0.17   | 0.22   | 0.3    |        |
| Efficacy rate         | 0.05   | 0.3    | 0.3    | 0.3    | 0.3    |        |
| Utility 2             | -0     | 0.2    | 0.1    | 0.1    | 0      |        |
| Utility 3             | 40.2   | 53.2   | 51.2   | 49.2   | 46     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 5.1    | 38     | 29.7   | 16.3   | 7.7    | 3.2    |
| U-BOIN                | 2.4    | 37.7   | 28.8   | 19.2   | 11.9   | 0      |
| EffTox                | 4      | 23     | 27     | 9      | 23     | 14     |
| AIDE one-stage (U2)   | 5.6    | 35.9   | 25     | 18.8   | 11.2   | 3.5    |
| AIDE one-stage (U3)   | 9      | 42.3   | 21.6   | 17.7   | 6.1    | 3.3    |
| AIDE two-stage (U2)   | 4.4    | 35.3   | 24.7   | 18.1   | 10.4   | 7.1    |
| AIDE two-stage (U3)   | 7.7    | 40.7   | 22.7   | 15.4   | 6.4    | 7.1    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 4.1    | 7.7    | 7.2    | 6.2    | 4.2    |        |
| U-BOIN                | 5.4    | 8.6    | 7.3    | 5.4    | 3.5    |        |
| EffTox                | 9      | 12.2   | 13.7   | 6.8    | 12.6   |        |
| AIDE one-stage (U2)   | 3.6    | 7      | 6.9    | 7.1    | 4.6    |        |
| AIDE one-stage (U3)   | 3.7    | 7.7    | 7.1    | 6.8    | 4      |        |
| AIDE two-stage (U2)   | 4.1    | 6.2    | 6.5    | 6.2    | 5.5    |        |
| AIDE two-stage (U3)   | 4.1    | 6.2    | 6.4    | 6.2    | 5.6    |        |
| Scenario 6            |        |        |        |        |        |        |
| DLT rate              | 0.29   | 0.34   | 0.38   | 0.43   | 0.52   |        |
| Efficacy rate         | 0.3    | 0.32   | 0.35   | 0.38   | 0.4    |        |
| Utility 2             | 0      | -0     | -0     | -0.1   | -0.1   |        |
| Utility 3             | 46.4   | 45.6   | 45.8   | 45.6   | 43.2   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 56.2   | 15     | 6.4    | 1.8    | 0.4    | 20.2   |
| U-BOIN                | 23.8   | 17     | 15.5   | 17.3   | 15.8   | 10.6   |
| EffTox                | 71     | 3      | 3      | 3      | 1      | 20     |
| AIDE one-stage (U2)   | 55.8   | 18.2   | 4.7    | 1.9    | 0.5    | 18.9   |
| AIDE one-stage (U3)   | 57.3   | 17.9   | 3.8    | 1.5    | 0.4    | 19.1   |
| AIDE two-stage (U2)   | 58.3   | 13.4   | 4      | 1.4    | 0.5    | 22.4   |
| AIDE two-stage (U3)   | 61     | 11.9   | 3.2    | 1      | 0.4    | 22.5   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 14.7   | 6.2    | 3.5    | 1.8    | 0.5    |        |
| U-BOIN                | 11.8   | 6.6    | 4.2    | 3      | 1.9    |        |
| EffTox                | 42.4   | 3      | 2.3    | 1.6    | 1.7    |        |
| AIDE one-stage (U2)   | 16.2   | 6.2    | 2.7    | 1.4    | 0.3    |        |
| AIDE one-stage (U3)   | 16.7   | 6.1    | 2.5    | 1.2    | 0.3    |        |
| AIDE two-stage (U2)   | 17.8   | 3.9    | 2.5    | 1.1    | 0.4    |        |
| AIDE two-stage (U3)   | 17.8   | 3.9    | 2.5    | 1.1    | 0.4    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 7            |        |        |        |        |        |        |
| DLT rate              | 0.16   | 0.2    | 0.26   | 0.31   | 0.36   |        |
| Efficacy rate         | 0.2    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2             | 0      | 0.3    | 0.2    | 0.2    | 0.1    |        |
| Utility 3             | 45.6   | 62     | 59.6   | 57.6   | 55.6   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 10.8   | 49.7   | 20.6   | 10.9   | 2.8    | 5.2    |
| U-BOIN                | 4.7    | 48.2   | 25.7   | 12.1   | 7.1    | 2.2    |
| EffTox                | 44     | 16     | 8      | 8      | 22     | 2      |
| AIDE one-stage (U2)   | 9.5    | 44.5   | 22     | 12     | 3.8    | 8.2    |
| AIDE one-stage (U3)   | 12     | 47     | 20.8   | 9.2    | 2.4    | 8.6    |
| AIDE two-stage (U2)   | 17     | 35.3   | 20.3   | 10     | 4.3    | 13.1   |
| AIDE two-stage (U3)   | 18.8   | 38.6   | 18     | 8.7    | 2.8    | 13.1   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 6.4    | 12.1   | 6.1    | 3.3    | 1.2    |        |
| U-BOIN                | 7.6    | 11.1   | 6.3    | 3.1    | 1.4    |        |
| EffTox                | 31.3   | 9.5    | 5.5    | 4      | 8.6    |        |
| AIDE one-stage (U2)   | 6.9    | 12.1   | 5.8    | 2.9    | 0.9    |        |
| AIDE one-stage (U3)   | 7.2    | 12.6   | 5.4    | 2.6    | 0.7    |        |
| AIDE two-stage (U2)   | 8.9    | 7      | 5.5    | 3.7    | 2.5    |        |
| AIDE two-stage (U3)   | 8.9    | 6.9    | 5.6    | 3.7    | 2.6    |        |
| Scenario 8            |        |        |        |        |        |        |
| DLT rate              | 0.32   | 0.37   | 0.42   | 0.48   | 0.52   |        |
| Efficacy rate         | 0.1    | 0.15   | 0.2    | 0.3    | 0.4    |        |
| Utility 2             | -0.2   | -0.2   | -0.2   | -0.2   | -0.1   |        |
| Utility 3             | 33.2   | 34.2   | 35.2   | 38.8   | 43.2   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 18.1   | 9.9    | 4.3    | 1.7    | 0      | 66     |
| U-BOIN                | 10.3   | 13.3   | 14.8   | 18.5   | 26     | 17.1   |
| EffTox                | 2      | 4      | 11     | 5      | 2      | 76     |
| AIDE one-stage (U2)   | 15.3   | 8.6    | 3      | 1.4    | 0.3    | 71.4   |
| AIDE one-stage (U3)   | 14     | 9.3    | 3.3    | 1      | 0.1    | 72.3   |
| AIDE two-stage (U2)   | 13.7   | 5.4    | 2.8    | 1.1    | 0.3    | 76.7   |
| AIDE two-stage (U3)   | 14.9   | 5.6    | 1.7    | 1      | 0.2    | 76.6   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 8.8    | 5.4    | 3.3    | 1.7    | 0.7    |        |
| U-BOIN                | 8.8    | 4.8    | 4.3    | 4.3    | 4.2    |        |
| EffTox                | 8.5    | 4.7    | 5.1    | 3.4    | 3.9    |        |
| AIDE one-stage (U2)   | 9.4    | 5.4    | 2.2    | 1.1    | 0.3    |        |
| AIDE one-stage (U3)   | 9.6    | 5.4    | 2.1    | 1.1    | 0.3    |        |
| AIDE two-stage (U2)   | 9.9    | 3.3    | 1.9    | 0.8    | 0.3    |        |
| AIDE two-stage (U3)   | 9.9    | 3.3    | 1.8    | 0.9    | 0.3    |        |



# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 9            |        |        |        |        |        |        |
| DLT rate              | 0.15   | 0.18   | 0.22   | 0.26   | 0.29   |        |
| Efficacy rate         | 0.05   | 0.08   | 0.2    | 0.25   | 0.5    |        |
| Utility 2             | -0.1   | -0.1   | -0     | -0     | 0.2    |        |
| Utility 3             | 37     | 37.6   | 43.2   | 44.6   | 58.4   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 11.5   | 8.8    | 20.8   | 14.7   | 23.9   | 20.3   |
| U-BOIN                | 2.3    | 6.1    | 16.5   | 18.8   | 55     | 1.3    |
| EffTox                | 0      | 1      | 8      | 5      | 67     | 20     |
| AIDE one-stage (U2)   | 8.9    | 6.1    | 18.1   | 15.9   | 26.4   | 24.6   |
| AIDE one-stage (U3)   | 12.2   | 8.6    | 17.5   | 15.1   | 22     | 24.6   |
| AIDE two-stage (U2)   | 4.7    | 5.1    | 15.3   | 13.9   | 25.7   | 35.3   |
| AIDE two-stage (U3)   | 9.4    | 6.9    | 13.9   | 11     | 23.3   | 35.5   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 5      | 5      | 5.9    | 5.2    | 5.8    |        |
| U-BOIN                | 6.5    | 5.9    | 6.2    | 5.2    | 6.1    |        |
| EffTox                | 4.9    | 3.9    | 6.7    | 5.3    | 30     |        |
| AIDE one-stage (U2)   | 5      | 4.9    | 5.5    | 5.2    | 5.1    |        |
| AIDE one-stage (U3)   | 5.2    | 5      | 5.5    | 5.2    | 4.8    |        |
| AIDE two-stage (U2)   | 5.6    | 3.8    | 5.2    | 4      | 5.2    |        |
| AIDE two-stage (U3)   | 5.6    | 3.8    | 5.1    | 4      | 5.2    |        |
| Scenario 10           |        |        |        |        |        |        |
| DLT rate              | 0.23   | 0.28   | 0.32   | 0.39   | 0.43   |        |
| Efficacy rate         | 0.05   | 0.4    | 0.42   | 0.45   | 0.5    |        |
| Utility 2             | -0.2   | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3             | 33.8   | 52.8   | 52.4   | 51.4   | 52.8   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 6.1    | 41.6   | 13.2   | 5.4    | 1.5    | 32.2   |
| U-BOIN                | 0.3    | 32.6   | 24.5   | 19.2   | 17.1   | 6.3    |
| EffTox                | 5      | 23     | 15     | 17     | 23     | 16     |
| AIDE one-stage (U2)   | 2.6    | 28     | 15.6   | 6.1    | 1.8    | 45.9   |
| AIDE one-stage (U3)   | 3.2    | 30.2   | 14.1   | 5.3    | 0.9    | 46.3   |
| AIDE two-stage (U2)   | 4.1    | 25.3   | 12     | 4      | 2.2    | 52.4   |
| AIDE two-stage (U3)   | 4.8    | 26.4   | 10.7   | 4      | 1.4    | 52.7   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 5.8    | 10.4   | 5      | 2.6    | 1      |        |
| U-BOIN                | 7.7    | 9.2    | 6.1    | 3.5    | 2      |        |
| EffTox                | 10.6   | 12.8   | 8.4    | 7.3    | 13.2   |        |
| AIDE one-stage (U2)   | 5.9    | 8.5    | 4.6    | 2.2    | 0.6    |        |
| AIDE one-stage (U3)   | 5.9    | 8.9    | 4.4    | 2      | 0.5    |        |
| AIDE two-stage (U2)   | 6.2    | 5.7    | 4      | 2.1    | 1.2    |        |
| AIDE two-stage (U3)   | 6.2    | 5.6    | 4      | 2.2    | 1.2    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 11           |        |        |        |        |        |        |
| DLT rate              | 0.22   | 0.32   | 0.44   | 0.59   | 0.67   |        |
| Efficacy rate         | 0.2    | 0.2    | 0.2    | 0.2    | 0.2    |        |
| Utility 2             | -0     | -0.1   | -0.2   | -0.4   | -0.5   |        |
| Utility 3             | 43.2   | 39.2   | 34.4   | 28.4   | 25.2   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 51.3   | 18.2   | 3.3    | 0.4    | 0      | 26.8   |
| U-BOIN                | 38.2   | 21.1   | 16.3   | 8.5    | 10.5   | 5.4    |
| EffTox                | 33     | 6      | 4      | 2      | 1      | 54     |
| AIDE one-stage (U2)   | 47.2   | 19.1   | 5.1    | 0.7    | 0      | 27.9   |
| AIDE one-stage (U3)   | 48.6   | 17.7   | 4.9    | 0.5    | 0.1    | 28.2   |
| AIDE two-stage (U2)   | 45.2   | 18.3   | 3.7    | 0.8    | 0      | 32     |
| AIDE two-stage (U3)   | 47.8   | 16.9   | 2.5    | 0.8    | 0      | 32     |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 13     | 7.5    | 3.9    | 1.6    | 0.4    |        |
| U-BOIN                | 11.7   | 6.9    | 4.7    | 3.2    | 2.6    |        |
| EffTox                | 26.2   | 4.3    | 3.3    | 1.6    | 0.8    |        |
| AIDE one-stage (U2)   | 13.7   | 7.3    | 3.3    | 1.4    | 0.2    |        |
| AIDE one-stage (U3)   | 14     | 7.1    | 3.1    | 1.3    | 0.2    |        |
| AIDE two-stage (U2)   | 14.4   | 5.7    | 3.4    | 1      | 0.2    |        |
| AIDE two-stage (U3)   | 14.3   | 5.7    | 3.5    | 1.1    | 0.2    |        |
| Scenario 12           |        |        |        |        |        |        |
| DLT rate              | 0.06   | 0.09   | 0.18   | 0.29   | 0.44   |        |
| Efficacy rate         | 0.05   | 0.1    | 0.12   | 0.4    | 0.6    |        |
| Utility 2             | -0     | 0      | -0.1   | 0.1    | 0.2    |        |
| Utility 3             | 40.6   | 42.4   | 40     | 52.4   | 58.4   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 13.3   | 17.5   | 13.4   | 34.7   | 11.2   | 9.9    |
| U-BOIN                | 5.4    | 9.5    | 8.6    | 37.4   | 38.7   | 0.4    |
| EffTox                | 0      | 0      | 15     | 48     | 24     | 13     |
| AIDE one-stage (U2)   | 10.9   | 8.8    | 7.9    | 42.2   | 24.3   | 5.9    |
| AIDE one-stage (U3)   | 13.8   | 11.1   | 7.4    | 40.3   | 21.3   | 6.1    |
| AIDE two-stage (U2)   | 10.6   | 9.2    | 7.7    | 36.3   | 24     | 12.2   |
| AIDE two-stage (U3)   | 15.4   | 9.6    | 6.7    | 34     | 22.4   | 11.9   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 4.6    | 5.7    | 6.1    | 8.5    | 4.3    |        |
| U-BOIN                | 5.1    | 6.2    | 6.5    | 7.2    | 5.1    |        |
| EffTox                | 5      | 4      | 11.4   | 15.6   | 18.8   |        |
| AIDE one-stage (U2)   | 4      | 4.7    | 5.2    | 9.4    | 5.5    |        |
| AIDE one-stage (U3)   | 4.1    | 4.7    | 5.1    | 9.6    | 5.2    |        |
| AIDE two-stage (U2)   | 4.2    | 4.5    | 5      | 7.4    | 7.3    |        |
| AIDE two-stage (U3)   | 4.2    | 4.4    | 5      | 7.4    | 7.4    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 13           |        |        |        |        |        |        |
| DLT rate              | 0.21   | 0.29   | 0.3    | 0.39   | 0.46   |        |
| Efficacy rate         | 0.05   | 0.3    | 0.32   | 0.35   | 0.4    |        |
| Utility 2             | -0.2   | 0      | 0      | -0     | -0.1   |        |
| Utility 3             | 34.6   | 46.4   | 47.2   | 45.4   | 45.6   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 7.2    | 34.6   | 18.3   | 5.1    | 1.8    | 33     |
| U-BOIN                | 1.9    | 24.2   | 26.4   | 22     | 22.4   | 3.1    |
| EffTox                | 8      | 18     | 17     | 16     | 15     | 27     |
| AIDE one-stage (U2)   | 4.2    | 28.2   | 15.2   | 6.8    | 1.7    | 43.9   |
| AIDE one-stage (U3)   | 6.1    | 28.6   | 14.1   | 5.5    | 1.2    | 44.5   |
| AIDE two-stage (U2)   | 6      | 25.1   | 13.6   | 4.7    | 2      | 48.6   |
| AIDE two-stage (U3)   | 7      | 26.8   | 12     | 4.3    | 1.4    | 48.5   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 6      | 9.1    | 5.6    | 3.1    | 1.3    |        |
| U-BOIN                | 7.8    | 7.7    | 6.4    | 4.5    | 2.9    |        |
| EffTox                | 9.7    | 11.1   | 9.3    | 6.6    | 11.1   |        |
| AIDE one-stage (U2)   | 6      | 7.8    | 4.7    | 2.8    | 0.9    |        |
| AIDE one-stage (U3)   | 6      | 8.1    | 4.7    | 2.7    | 0.8    |        |
| AIDE two-stage (U2)   | 6.2    | 5.9    | 4.5    | 2.4    | 1.3    |        |
| AIDE two-stage (U3)   | 6.2    | 5.9    | 4.4    | 2.4    | 1.2    |        |
| Scenario 14           |        |        |        |        |        |        |
| DLT rate              | 0.09   | 0.15   | 0.23   | 0.28   | 0.39   |        |
| Efficacy rate         | 0.05   | 0.08   | 0.12   | 0.3    | 0.4    |        |
| Utility 2             | -0     | -0.1   | -0.1   | 0      | 0      |        |
| Utility 3             | 39.4   | 38.8   | 38     | 46.8   | 48.4   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 16.3   | 14.7   | 15     | 25.4   | 8.7    | 19.9   |
| U-BOIN                | 6.2    | 8.8    | 10.5   | 36.7   | 36.9   | 0.9    |
| EffTox                | 0      | 0      | 14     | 24     | 30     | 33     |
| AIDE one-stage (U2)   | 13.5   | 10.8   | 12.1   | 32.1   | 16     | 15.5   |
| AIDE one-stage (U3)   | 18     | 13.3   | 11.6   | 29.7   | 11.4   | 16     |
| AIDE two-stage (U2)   | 12.4   | 8.7    | 10.2   | 28.8   | 16.3   | 23.6   |
| AIDE two-stage (U3)   | 18.3   | 12.1   | 8.4    | 24.5   | 12.9   | 23.8   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 5.2    | 6.1    | 6      | 6.8    | 4      |        |
| U-BOIN                | 6.2    | 6.5    | 5.9    | 6.4    | 5.1    |        |
| EffTox                | 4.9    | 3.9    | 9.6    | 9.9    | 17.4   |        |
| AIDE one-stage (U2)   | 4.8    | 5.2    | 5.3    | 7.7    | 4.2    |        |
| AIDE one-stage (U3)   | 5      | 5.4    | 5.1    | 7.7    | 4      |        |
| AIDE two-stage (U2)   | 5.1    | 4.7    | 5.4    | 6.2    | 5.1    |        |
| AIDE two-stage (U3)   | 5      | 4.7    | 5.5    | 6.3    | 5.1    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 15           |        |        |        |        |        |        |
| DLT rate              | 0.13   | 0.2    | 0.32   | 0.39   | 0.46   |        |
| Efficacy rate         | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 2             | -0     | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3             | 40.8   | 38     | 33.2   | 30.4   | 27.6   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 35.6   | 20.2   | 8.3    | 3      | 0.6    | 32.3   |
| U-BOIN                | 28.9   | 18.6   | 16     | 16.7   | 18.2   | 1.6    |
| EffTox                | 1      | 1      | 3      | 2      | 3      | 91     |
| AIDE one-stage (U2)   | 32.4   | 16.4   | 14.1   | 6.7    | 1.4    | 29     |
| AIDE one-stage (U3)   | 33.1   | 18.1   | 12.8   | 5.4    | 1.1    | 29.5   |
| AIDE two-stage (U2)   | 30.3   | 15.2   | 8.2    | 3.3    | 0.9    | 42.1   |
| AIDE two-stage (U3)   | 37.2   | 15     | 5      | 1.3    | 0.4    | 41.1   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 8.2    | 7.4    | 5.8    | 3.7    | 1.9    |        |
| U-BOIN                | 8.7    | 7.3    | 6      | 4.3    | 3.6    |        |
| EffTox                | 8      | 3.9    | 4.5    | 3.3    | 3.2    |        |
| AIDE one-stage (U2)   | 7.7    | 7.1    | 5.4    | 4      | 1.5    |        |
| AIDE one-stage (U3)   | 8      | 7.1    | 5.3    | 3.9    | 1.5    |        |
| AIDE two-stage (U2)   | 8.1    | 6      | 5.5    | 3.4    | 1.7    |        |
| AIDE two-stage (U3)   | 8.2    | 5.9    | 5.6    | 3.4    | 1.7    |        |
| Scenario 16           |        |        |        |        |        |        |
| DLT rate              | 0.02   | 0.06   | 0.09   | 0.14   | 0.26   |        |
| Efficacy rate         | 0.2    | 0.2    | 0.2    | 0.2    | 0.2    |        |
| Utility 2             | 0.2    | 0.1    | 0.1    | 0.1    | -0.1   |        |
| Utility 3             | 51.2   | 49.6   | 48.4   | 46.4   | 41.6   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 34     | 24.5   | 23.6   | 11.8   | 5.5    | 0.6    |
| U-BOIN                | 28.5   | 23.5   | 20.9   | 17.5   | 9.6    | 0      |
| EffTox                | 32     | 5      | 13     | 6      | 10     | 35     |
| AIDE one-stage (U2)   | 40.2   | 24.9   | 14.7   | 12.9   | 7.3    | 0      |
| AIDE one-stage (U3)   | 45.5   | 25.5   | 13.4   | 10.8   | 4.6    | 0.2    |
| AIDE two-stage (U2)   | 31.1   | 24.9   | 16.8   | 14.1   | 12.5   | 0.6    |
| AIDE two-stage (U3)   | 38.8   | 26.8   | 16.9   | 10.3   | 6.6    | 0.6    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 4.9    | 5.7    | 7      | 7.2    | 5.2    |        |
| U-BOIN                | 6.3    | 6      | 6.2    | 6.5    | 5.4    |        |
| EffTox                | 24.2   | 4.4    | 7      | 5.1    | 6.9    |        |
| AIDE one-stage (U2)   | 4.5    | 4.9    | 6      | 8.2    | 6.4    |        |
| AIDE one-stage (U3)   | 4.7    | 5.1    | 6.1    | 8.2    | 5.8    |        |
| AIDE two-stage (U2)   | 5.5    | 4.9    | 5.1    | 7.1    | 7.3    |        |
| AIDE two-stage (U3)   | 5.5    | 4.9    | 5.2    | 7.1    | 7.2    |        |



# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 17           |        |        |        |        |        |        |
| DLT rate              | 0.11   | 0.23   | 0.39   | 0.58   | 0.76   |        |
| Efficacy rate         | 0.3    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2             | 0.2    | 0.3    | 0.1    | -0.1   | -0.3   |        |
| Utility 3             | 53.6   | 60.8   | 54.4   | 46.8   | 39.6   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 30.2   | 56.8   | 10.6   | 0.6    | 0      | 1.8    |
| U-BOIN                | 24.8   | 53     | 15.2   | 4      | 2.6    | 0.4    |
| EffTox                | 79     | 15     | 3      | 0      | 0      | 3      |
| AIDE one-stage (U2)   | 18.7   | 58.5   | 17.2   | 2.4    | 0.1    | 3.1    |
| AIDE one-stage (U3)   | 21.6   | 60.1   | 13.5   | 1.9    | 0      | 2.9    |
| AIDE two-stage (U2)   | 28.9   | 48.3   | 16.9   | 2.3    | 0      | 3.6    |
| AIDE two-stage (U3)   | 32     | 48.4   | 14.1   | 1.8    | 0.1    | 3.6    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 9.8    | 13.6   | 4.9    | 1.2    | 0.2    |        |
| U-BOIN                | 10.3   | 12.1   | 5.4    | 1.7    | 0.4    |        |
| EffTox                | 46.5   | 7.6    | 3.1    | 0.9    | 0.3    |        |
| AIDE one-stage (U2)   | 8.8    | 14     | 5.3    | 1.3    | 0.2    |        |
| AIDE one-stage (U3)   | 9.3    | 14.3   | 4.7    | 1.1    | 0.1    |        |
| AIDE two-stage (U2)   | 9.6    | 9.7    | 7.3    | 2.3    | 0.5    |        |
| AIDE two-stage (U3)   | 9.6    | 9.8    | 7.2    | 2.4    | 0.4    |        |
| Scenario 18           |        |        |        |        |        |        |
| DLT rate              | 0.02   | 0.04   | 0.1    | 0.2    | 0.35   |        |
| Efficacy rate         | 0.05   | 0.1    | 0.15   | 0.2    | 0.5    |        |
| Utility 2             | 0      | 0.1    | 0.1    | 0      | 0.1    |        |
| Utility 3             | 42.2   | 44.4   | 45     | 44     | 56     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 14.1   | 18.9   | 20.5   | 15.4   | 27.9   | 3.2    |
| U-BOIN                | 5.8    | 11.6   | 19.2   | 16.4   | 46.8   | 0.2    |
| EffTox                | 0      | 0      | 14     | 22     | 45     | 19     |
| AIDE one-stage (U2)   | 10.5   | 9.9    | 8.8    | 14.5   | 55.4   | 0.9    |
| AIDE one-stage (U3)   | 15.8   | 13.7   | 9.4    | 13     | 47.4   | 0.7    |
| AIDE two-stage (U2)   | 9.8    | 8.4    | 9.4    | 13.8   | 56.3   | 2.3    |
| AIDE two-stage (U3)   | 15.5   | 10.7   | 9      | 11.9   | 50.8   | 2.1    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 3.9    | 5      | 6      | 6.7    | 8.1    |        |
| U-BOIN                | 4.2    | 5.4    | 7.1    | 7      | 6.6    |        |
| EffTox                | 4.6    | 3.6    | 9.6    | 11.9   | 23.2   |        |
| AIDE one-stage (U2)   | 3.4    | 3.9    | 5.1    | 6.8    | 10.6   |        |
| AIDE one-stage (U3)   | 3.4    | 4.1    | 5.2    | 6.8    | 10.2   |        |
| AIDE two-stage (U2)   | 3.6    | 3.8    | 4.6    | 5.6    | 12.1   |        |
| AIDE two-stage (U3)   | 3.6    | 3.8    | 4.6    | 5.5    | 12.2   |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 19           |        |        |        |        |        |        |
| DLT rate              | 0.11   | 0.26   | 0.39   | 0.6    | 0.77   |        |
| Efficacy rate         | 0.08   | 0.3    | 0.32   | 0.35   | 0.4    |        |
| Utility 2             | -0     | 0      | -0.1   | -0.2   | -0.4   |        |
| Utility 3             | 40.4   | 47.6   | 43.6   | 37     | 33.2   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 11.5   | 53     | 12.2   | 0.5    | 0      | 22.8   |
| U-BOIN                | 10.5   | 41.7   | 24.3   | 12.7   | 9      | 1.8    |
| EffTox                | 10     | 37     | 13     | 1      | 1      | 38     |
| AIDE one-stage (U2)   | 8.5    | 45     | 18.2   | 2.8    | 0.1    | 25.4   |
| AIDE one-stage (U3)   | 9.6    | 45.7   | 16.2   | 2.1    | 0.1    | 26.3   |
| AIDE two-stage (U2)   | 11.9   | 44.9   | 16     | 2.2    | 0.1    | 24.9   |
| AIDE two-stage (U3)   | 14.9   | 44.9   | 13.2   | 1.7    | 0.1    | 25.2   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 7.4    | 12.1   | 5.6    | 1.8    | 0.3    |        |
| U-BOIN                | 7.8    | 10     | 6.4    | 3.5    | 2.1    |        |
| EffTox                | 13     | 16.5   | 10.2   | 3.1    | 1.4    |        |
| AIDE one-stage (U2)   | 6.1    | 11     | 6.3    | 2.3    | 0.4    |        |
| AIDE one-stage (U3)   | 6.2    | 11.2   | 6.1    | 2.1    | 0.4    |        |
| AIDE two-stage (U2)   | 6.3    | 9.8    | 6.9    | 2.2    | 0.4    |        |
| AIDE two-stage (U3)   | 6.3    | 9.8    | 6.9    | 2.2    | 0.4    |        |
| Scenario 20           |        |        |        |        |        |        |
| DLT rate              | 0.07   | 0.14   | 0.31   | 0.41   | 0.52   |        |
| Efficacy rate         | 0.05   | 0.25   | 0.25   | 0.25   | 0.25   |        |
| Utility 2             | -0     | 0.1    | -0.1   | -0.2   | -0.3   |        |
| Utility 3             | 40.2   | 49.4   | 42.6   | 38.6   | 34.2   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 7.8    | 58.3   | 20.5   | 3.7    | 0.6    | 9.1    |
| U-BOIN                | 6.6    | 45     | 19.6   | 15.5   | 13.1   | 0.2    |
| EffTox                | 4      | 20     | 26     | 3      | 3      | 45     |
| AIDE one-stage (U2)   | 9.1    | 42.5   | 26.7   | 11.9   | 1.8    | 8      |
| AIDE one-stage (U3)   | 11.8   | 44.6   | 23.5   | 10.4   | 1.4    | 8.3    |
| AIDE two-stage (U2)   | 10.1   | 41.6   | 25.7   | 9.5    | 1.4    | 11.7   |
| AIDE two-stage (U3)   | 16.1   | 42.2   | 21.4   | 7.4    | 0.9    | 12     |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 5.4    | 10.9   | 7.3    | 4.1    | 1.6    |        |
| U-BOIN                | 5.9    | 10.7   | 6.9    | 4.2    | 2.4    |        |
| EffTox                | 8.2    | 12     | 14.1   | 4.8    | 3.3    |        |
| AIDE one-stage (U2)   | 4.3    | 9.5    | 7.9    | 5.1    | 1.8    |        |
| AIDE one-stage (U3)   | 4.5    | 10     | 7.7    | 4.8    | 1.6    |        |
| AIDE two-stage (U2)   | 4.7    | 8.3    | 8.2    | 4.7    | 2.2    |        |
| AIDE two-stage (U3)   | 4.7    | 8.3    | 8.2    | 4.7    | 2.3    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 21           |        |        |        |        |        |        |
| DLT rate              | 0.37   | 0.56   | 0.69   | 0.83   | 0.88   |        |
| Efficacy rate         | 0.5    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2             | 0.1    | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3             | 55.2   | 47.6   | 42.4   | 36.8   | 34.8   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 59.6   | 0.9    | 0      | 0      | 0      | 39.5   |
| U-BOIN                | 53.3   | 6.6    | 5.8    | 5.2    | 4      | 25.1   |
| EffTox                | 58     | 0      | 0      | 0      | 0      | 42     |
| AIDE one-stage (U2)   | 58.5   | 1.8    | 0      | 0      | 0      | 39.7   |
| AIDE one-stage (U3)   | 58.3   | 1.7    | 0      | 0      | 0      | 40     |
| AIDE two-stage (U2)   | 60.9   | 1.5    | 0      | 0      | 0      | 37.6   |
| AIDE two-stage (U3)   | 61.4   | 1.4    | 0      | 0      | 0      | 37.2   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 20.1   | 2.7    | 0.4    | 0      | 0      |        |
| U-BOIN                | 17.3   | 3.9    | 1.5    | 0.9    | 0.7    |        |
| EffTox                | 42.7   | 0.6    | 0.2    | 0      | 0      |        |
| AIDE one-stage (U2)   | 20.3   | 2.6    | 0.2    | 0      | 0      |        |
| AIDE one-stage (U3)   | 20.4   | 2.5    | 0.2    | 0      | 0      |        |
| AIDE two-stage (U2)   | 20.9   | 2.1    | 0.6    | 0      | 0      |        |
| AIDE two-stage (U3)   | 20.9   | 2.1    | 0.6    | 0      | 0      |        |
| Scenario 22           |        |        |        |        |        |        |
| DLT rate              | 0.07   | 0.27   | 0.5    | 0.56   | 0.78   |        |
| Efficacy rate         | 0.08   | 0.25   | 0.25   | 0.5    | 0.5    |        |
| Utility 2             | 0      | -0     | -0.2   | -0.1   | -0.3   |        |
| Utility 3             | 42     | 44.2   | 35     | 47.6   | 38.8   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 15.4   | 53.9   | 6.3    | 0.4    | 0      | 24     |
| U-BOIN                | 13.7   | 34.2   | 10.5   | 26.3   | 13.5   | 1.8    |
| EffTox                | 4      | 24     | 9      | 2      | 0      | 61     |
| AIDE one-stage (U2)   | 11.7   | 44.8   | 12.2   | 4.3    | 0.1    | 26.9   |
| AIDE one-stage (U3)   | 12.9   | 45.1   | 11.3   | 3.3    | 0      | 27.4   |
| AIDE two-stage (U2)   | 17.9   | 42.8   | 10.5   | 1.5    | 0      | 27.3   |
| AIDE two-stage (U3)   | 20.4   | 42.1   | 8.8    | 1.5    | 0      | 27.2   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 8      | 12.3   | 4.9    | 1.8    | 0.3    |        |
| U-BOIN                | 8.2    | 9.6    | 5      | 4.7    | 2.5    |        |
| EffTox                | 10.3   | 12.7   | 8.6    | 2.6    | 0.9    |        |
| AIDE one-stage (U2)   | 6.6    | 11.5   | 5.5    | 2.2    | 0.2    |        |
| AIDE one-stage (U3)   | 6.9    | 11.6   | 5.3    | 2      | 0.2    |        |
| AIDE two-stage (U2)   | 6.6    | 11.1   | 6.4    | 1.7    | 0.3    |        |
| AIDE two-stage (U3)   | 6.6    | 11.2   | 6.4    | 1.7    | 0.3    |        |

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda\_T = 0.3; U3: Type = 3, Lambda\_T = 1.

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 23           |        |        |        |        |        |        |
| DLT rate              | 0.32   | 0.55   | 0.8    | 0.94   | 0.97   |        |
| Efficacy rate         | 0.05   | 0.05   | 0.05   | 0.05   | 0.05   |        |
| Utility 2             | -0.3   | -0.5   | -0.8   | -0.9   | -0.9   |        |
| Utility 3             | 30.2   | 21     | 11     | 5.4    | 4.2    |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 3.1    | 0.6    | 0      | 0      | 0      | 96.3   |
| U-BOIN                | 7      | 6.6    | 2.3    | 2.3    | 1      | 80.8   |
| EffTox                | 0      | 1      | 6      | 1      | 0      | 92     |
| AIDE one-stage (U2)   | 1.9    | 0.3    | 0.1    | 0      | 0      | 97.7   |
| AIDE one-stage (U3)   | 1.9    | 0.4    | 0.2    | 0      | 0      | 97.5   |
| AIDE two-stage (U2)   | 1.5    | 0.5    | 0      | 0      | 0      | 98     |
| AIDE two-stage (U3)   | 1.5    | 0.4    | 0      | 0      | 0      | 98.1   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 8.1    | 4      | 1.2    | 0.1    | 0      |        |
| U-BOIN                | 9.8    | 5.1    | 3.7    | 2.8    | 2.7    |        |
| EffTox                | 4.5    | 3.1    | 1.9    | 0.3    | 0      |        |
| AIDE one-stage (U2)   | 7.7    | 3.9    | 0.7    | 0.1    | 0      |        |
| AIDE one-stage (U3)   | 7.7    | 3.9    | 0.7    | 0.1    | 0      |        |
| AIDE two-stage (U2)   | 8      | 2.1    | 0.6    | 0      | 0      |        |
| AIDE two-stage (U3)   | 8      | 2.2    | 0.6    | 0      | 0      |        |
| Scenario 24           |        |        |        |        |        |        |
| DLT rate              | 0.01   | 0.02   | 0.05   | 0.1    | 0.25   |        |
| Efficacy rate         | 0.05   | 0.08   | 0.2    | 0.5    | 0.5    |        |
| Utility 2             | 0      | 0.1    | 0.1    | 0.4    | 0.2    |        |
| Utility 3             | 42.6   | 44     | 50     | 66     | 60     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 2      | 2.6    | 6.6    | 64.3   | 24     | 0.5    |
| U-BOIN                | 0.6    | 1.7    | 8.6    | 60.7   | 28.4   | 0      |
| EffTox                | 0      | 0      | 6      | 21     | 69     | 5      |
| AIDE one-stage (U2)   | 2.5    | 2.1    | 3.9    | 53     | 38.3   | 0.2    |
| AIDE one-stage (U3)   | 3.9    | 3.1    | 5.3    | 56.5   | 31     | 0.2    |
| AIDE two-stage (U2)   | 1.3    | 0.9    | 5.7    | 45.7   | 45.9   | 0.5    |
| AIDE two-stage (U3)   | 2.5    | 2.4    | 7.6    | 46     | 41     | 0.5    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 3.1    | 3.4    | 4.4    | 12.9   | 6      |        |
| U-BOIN                | 3.4    | 3.9    | 5.1    | 9.7    | 8.1    |        |
| EffTox                | 4.5    | 3.5    | 6.3    | 10.2   | 33.4   |        |
| AIDE one-stage (U2)   | 3.1    | 3.1    | 3.9    | 12.4   | 7.4    |        |
| AIDE one-stage (U3)   | 3.1    | 3.2    | 4      | 12.9   | 6.7    |        |
| AIDE two-stage (U2)   | 3.3    | 3.2    | 3.7    | 8.4    | 11.3   |        |
| AIDE two-stage (U3)   | 3.3    | 3.2    | 3.8    | 8.4    | 11.3   |        |



# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 25           |        |        |        |        |        |        |
| DLT rate              | 0.09   | 0.18   | 0.37   | 0.63   | 0.77   |        |
| Efficacy rate         | 0.2    | 0.25   | 0.4    | 0.45   | 0.5    |        |
| Utility 2             | 0.1    | 0.1    | 0      | -0.2   | -0.3   |        |
| Utility 3             | 48.4   | 47.8   | 49.2   | 41.8   | 39.2   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 37     | 35.4   | 20.4   | 0.4    | 0      | 6.8    |
| U-BOIN                | 29.2   | 25.8   | 25.8   | 9.7    | 8.8    | 0.7    |
| EffTox                | 39     | 23     | 15     | 2      | 1      | 21     |
| AIDE one-stage (U2)   | 24.5   | 27.2   | 36.5   | 5.6    | 0.1    | 6.1    |
| AIDE one-stage (U3)   | 29.3   | 28.4   | 31     | 4.7    | 0      | 6.6    |
| AIDE two-stage (U2)   | 28.9   | 25.3   | 32.5   | 5.5    | 0      | 7.8    |
| AIDE two-stage (U3)   | 33.6   | 26.3   | 28.4   | 4      | 0      | 7.7    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 9.3    | 10     | 7.8    | 2      | 0.3    |        |
| U-BOIN                | 9.2    | 9      | 7.5    | 3      | 1.2    |        |
| EffTox                | 26.8   | 11.5   | 9.6    | 2.6    | 0.9    |        |
| AIDE one-stage (U2)   | 8.3    | 8.8    | 8.6    | 2.9    | 0.4    |        |
| AIDE one-stage (U3)   | 8.7    | 9      | 8.5    | 2.6    | 0.3    |        |
| AIDE two-stage (U2)   | 8.5    | 7.8    | 9.1    | 2.8    | 0.5    |        |
| AIDE two-stage (U3)   | 8.5    | 7.9    | 9.1    | 2.8    | 0.5    |        |
| Scenario 26           |        |        |        |        |        |        |
| DLT rate              | 0.15   | 0.22   | 0.3    | 0.62   | 0.8    |        |
| Efficacy rate         | 0.1    | 0.2    | 0.35   | 0.45   | 0.55   |        |
| Utility 2             | -0.1   | -0     | 0.1    | -0.2   | -0.2   |        |
| Utility 3             | 40     | 43.2   | 49     | 42.2   | 41     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 16.5   | 28     | 34.1   | 1.4    | 0      | 20     |
| U-BOIN                | 9.9    | 22     | 42.3   | 13.4   | 10.6   | 1.8    |
| EffTox                | 5      | 24     | 34     | 4      | 1      | 32     |
| AIDE one-stage (U2)   | 14.9   | 22.1   | 30.7   | 7.4    | 0.1    | 24.8   |
| AIDE one-stage (U3)   | 16.6   | 24.1   | 28.3   | 6.2    | 0.1    | 24.7   |
| AIDE two-stage (U2)   | 14.6   | 20.4   | 26.9   | 6.5    | 0.1    | 31.5   |
| AIDE two-stage (U3)   | 18.4   | 21.9   | 23     | 5.4    | 0      | 31.3   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 7.3    | 8.6    | 8.4    | 2.8    | 0.4    |        |
| U-BOIN                | 7.7    | 7.4    | 8.4    | 4.2    | 2.1    |        |
| EffTox                | 11.3   | 12.3   | 16.5   | 4.9    | 2      |        |
| AIDE one-stage (U2)   | 6.7    | 7.9    | 8      | 3.1    | 0.4    |        |
| AIDE one-stage (U3)   | 6.9    | 8      | 7.8    | 3      | 0.4    |        |
| AIDE two-stage (U2)   | 7.2    | 6.3    | 7.8    | 2.9    | 0.5    |        |
| AIDE two-stage (U3)   | 7.2    | 6.2    | 7.9    | 2.9    | 0.5    |        |

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda\_T = 0.3; U3: Type = 3, Lambda\_T = 1.

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 27           |        |        |        |        |        |        |
| DLT rate              | 0.11   | 0.17   | 0.3    | 0.5    | 0.62   |        |
| Efficacy rate         | 0.1    | 0.2    | 0.4    | 0.42   | 0.45   |        |
| Utility 2             | -0     | 0      | 0.1    | -0.1   | -0.2   |        |
| Utility 3             | 41.6   | 45.2   | 52     | 45.2   | 42.2   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 16.1   | 30.1   | 39.8   | 2.2    | 0.4    | 11.4   |
| U-BOIN                | 7.5    | 18     | 41.5   | 18.9   | 13.9   | 0.2    |
| EffTox                | 5      | 26     | 36     | 10     | 3      | 20     |
| AIDE one-stage (U2)   | 11.9   | 18.7   | 44.1   | 11.2   | 0.5    | 13.6   |
| AIDE one-stage (U3)   | 14.8   | 20.3   | 40.2   | 10.6   | 0.4    | 13.7   |
| AIDE two-stage (U2)   | 13.2   | 18.4   | 41     | 9.6    | 0.7    | 17.1   |
| AIDE two-stage (U3)   | 16.6   | 19.6   | 37.5   | 8.4    | 0.4    | 17.5   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 6.3    | 8.4    | 9.7    | 3.5    | 0.9    |        |
| U-BOIN                | 6.9    | 8.1    | 8.7    | 4.4    | 1.9    |        |
| EffTox                | 11.7   | 12.6   | 16.6   | 6.8    | 3.7    |        |
| AIDE one-stage (U2)   | 5.5    | 7.4    | 9.9    | 4      | 0.9    |        |
| AIDE one-stage (U3)   | 5.7    | 7.6    | 9.8    | 3.8    | 0.7    |        |
| AIDE two-stage (U2)   | 6      | 6      | 9.6    | 4      | 1.4    |        |
| AIDE two-stage (U3)   | 6      | 6      | 9.5    | 4      | 1.4    |        |
| Scenario 28           |        |        |        |        |        |        |
| DLT rate              | 0.2    | 0.3    | 0.5    | 0.6    | 0.77   |        |
| Efficacy rate         | 0.05   | 0.3    | 0.32   | 0.35   | 0.4    |        |
| Utility 2             | -0.1   | 0      | -0.2   | -0.2   | -0.4   |        |
| Utility 3             | 35     | 46     | 39.2   | 37     | 33.2   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 8.1    | 43.6   | 3.8    | 0.1    | 0      | 44.4   |
| U-BOIN                | 5      | 43.2   | 18.2   | 15.1   | 11.6   | 6.9    |
| EffTox                | 7      | 26     | 5      | 2      | 1      | 60     |
| AIDE one-stage (U2)   | 5.7    | 32.4   | 7.2    | 0.4    | 0      | 54.3   |
| AIDE one-stage (U3)   | 5.3    | 32.8   | 6.8    | 0.5    | 0      | 54.6   |
| AIDE two-stage (U2)   | 8.6    | 29.7   | 5.3    | 0.5    | 0      | 55.9   |
| AIDE two-stage (U3)   | 10.3   | 28.9   | 4.6    | 0.4    | 0      | 55.8   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 7.1    | 11.3   | 3.9    | 1.2    | 0.2    |        |
| U-BOIN                | 8      | 9.4    | 5.6    | 3.6    | 2.5    |        |
| EffTox                | 10.6   | 13.7   | 6.3    | 2.5    | 1.2    |        |
| AIDE one-stage (U2)   | 6.5    | 9.6    | 3.8    | 1.2    | 0.2    |        |
| AIDE one-stage (U3)   | 6.6    | 9.7    | 3.7    | 1.1    | 0.1    |        |
| AIDE two-stage (U2)   | 6.7    | 7.9    | 4.1    | 0.9    | 0.1    |        |
| AIDE two-stage (U3)   | 6.6    | 7.9    | 4.2    | 0.9    | 0.1    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 29           |        |        |        |        |        |        |
| DLT rate              | 0.19   | 0.3    | 0.56   | 0.74   | 0.86   |        |
| Efficacy rate         | 0.4    | 0.4    | 0.4    | 0.4    | 0.4    |        |
| Utility 2             | 0.2    | 0.1    | -0.2   | -0.3   | -0.5   |        |
| Utility 3             | 56.4   | 52     | 41.6   | 34.4   | 29.6   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 71.9   | 23.5   | 0.9    | 0      | 0      | 3.7    |
| U-BOIN                | 60.3   | 24.9   | 6.1    | 3.6    | 2.5    | 2.6    |
| EffTox                | 91     | 2      | 0      | 0      | 0      | 8      |
| AIDE one-stage (U2)   | 61.1   | 29.4   | 4.6    | 0.2    | 0      | 4.7    |
| AIDE one-stage (U3)   | 65.3   | 27.1   | 2.8    | 0.2    | 0      | 4.6    |
| AIDE two-stage (U2)   | 65.4   | 25.1   | 4.3    | 0.3    | 0      | 4.9    |
| AIDE two-stage (U3)   | 68.1   | 23.5   | 3.1    | 0.2    | 0      | 5.1    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 17     | 8.8    | 2.9    | 0.6    | 0      |        |
| U-BOIN                | 16.4   | 8.3    | 3      | 1      | 0.7    |        |
| EffTox                | 53.1   | 2.1    | 0.8    | 0.2    | 0      |        |
| AIDE one-stage (U2)   | 17.2   | 8.7    | 2.7    | 0.6    | 0      |        |
| AIDE one-stage (U3)   | 18     | 8.5    | 2.2    | 0.5    | 0      |        |
| AIDE two-stage (U2)   | 17.7   | 7.3    | 3.5    | 0.6    | 0      |        |
| AIDE two-stage (U3)   | 17.7   | 7.2    | 3.6    | 0.5    | 0      |        |
| Scenario 30           |        |        |        |        |        |        |
| DLT rate              | 0.05   | 0.08   | 0.17   | 0.27   | 0.54   |        |
| Efficacy rate         | 0.12   | 0.15   | 0.2    | 0.4    | 0.5    |        |
| Utility 2             | 0.1    | 0.1    | 0      | 0.1    | -0     |        |
| Utility 3             | 45.2   | 45.8   | 45.2   | 53.2   | 48.4   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 22.2   | 20.1   | 19.3   | 32.7   | 2.1    | 3.6    |
| U-BOIN                | 13.1   | 14.6   | 16.5   | 37.8   | 18     | 0      |
| EffTox                | 4      | 7      | 40     | 31     | 7      | 12     |
| AIDE one-stage (U2)   | 15.9   | 11.2   | 12.2   | 43.8   | 14.3   | 2.6    |
| AIDE one-stage (U3)   | 23.9   | 14.3   | 10.7   | 38.1   | 10.3   | 2.7    |
| AIDE two-stage (U2)   | 18.2   | 12.6   | 14.3   | 37.1   | 12.6   | 5.2    |
| AIDE two-stage (U3)   | 24     | 13.8   | 12.1   | 33.8   | 11.1   | 5.2    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 5      | 5.9    | 6.6    | 8.8    | 3.3    |        |
| U-BOIN                | 6.1    | 6.7    | 7      | 7.1    | 3.4    |        |
| EffTox                | 10.8   | 6      | 17.6   | 12.9   | 7.8    |        |
| AIDE one-stage (U2)   | 4.1    | 4.7    | 5.8    | 10.3   | 4.6    |        |
| AIDE one-stage (U3)   | 4.2    | 4.8    | 5.8    | 10.3   | 4.3    |        |
| AIDE two-stage (U2)   | 4.6    | 4.5    | 5.7    | 8.1    | 6.4    |        |
| AIDE two-stage (U3)   | 4.6    | 4.5    | 5.7    | 8.2    | 6.3    |        |

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda\_T = 0.3; U3: Type = 3, Lambda\_T = 1.

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 31           |        |        |        |        |        |        |
| DLT rate              | 0.04   | 0.06   | 0.1    | 0.2    | 0.6    |        |
| Efficacy rate         | 0.2    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2             | 0.2    | 0.4    | 0.4    | 0.3    | -0.1   |        |
| Utility 3             | 50.4   | 67.6   | 66     | 62     | 46     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 2.8    | 42.8   | 34.4   | 18.3   | 1.3    | 0.4    |
| U-BOIN                | 4.1    | 40.2   | 34.6   | 19.7   | 1.4    | 0      |
| EffTox                | 30     | 18     | 29     | 20     | 3      | 0      |
| AIDE one-stage (U2)   | 4.8    | 33.8   | 28.2   | 25.6   | 7.5    | 0.1    |
| AIDE one-stage (U3)   | 7.2    | 40.3   | 27.3   | 20.2   | 4.9    | 0.1    |
| AIDE two-stage (U2)   | 7.6    | 33.6   | 25.6   | 25.4   | 7.2    | 0.6    |
| AIDE two-stage (U3)   | 9.8    | 41.4   | 23.4   | 20.6   | 4.2    | 0.6    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 3.9    | 12.5   | 7.7    | 4.6    | 1.2    |        |
| U-BOIN                | 4.5    | 7.7    | 7.8    | 7.5    | 2.9    |        |
| EffTox                | 25.2   | 9.6    | 13     | 9.6    | 2.6    |        |
| AIDE one-stage (U2)   | 3.8    | 10.9   | 7.4    | 5.8    | 2.1    |        |
| AIDE one-stage (U3)   | 3.9    | 11.9   | 7.2    | 5.3    | 1.6    |        |
| AIDE two-stage (U2)   | 4      | 5.7    | 6.4    | 7.7    | 6      |        |
| AIDE two-stage (U3)   | 4      | 5.7    | 6.4    | 7.7    | 6      |        |
| Scenario 32           |        |        |        |        |        |        |
| DLT rate              | 0.02   | 0.05   | 0.11   | 0.3    | 0.36   |        |
| Efficacy rate         | 0.08   | 0.15   | 0.3    | 0.3    | 0.3    |        |
| Utility 2             | 0.1    | 0.1    | 0.2    | 0      | -0.1   |        |
| Utility 3             | 44     | 47     | 53.6   | 46     | 43.6   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 13.3   | 21     | 45.7   | 15.6   | 2.9    | 1.5    |
| U-BOIN                | 7.3    | 16.7   | 47.1   | 16.5   | 12.4   | 0      |
| EffTox                | 1      | 7      | 53     | 10     | 9      | 20     |
| AIDE one-stage (U2)   | 13.4   | 14.8   | 34.5   | 25.3   | 11.6   | 0.4    |
| AIDE one-stage (U3)   | 19.4   | 17.2   | 31.9   | 23.1   | 8      | 0.4    |
| AIDE two-stage (U2)   | 8.8    | 14.1   | 34.1   | 27.2   | 14.4   | 1.4    |
| AIDE two-stage (U3)   | 15.3   | 17.6   | 33.8   | 22.6   | 9.2    | 1.5    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 4.1    | 5.7    | 9.4    | 6.9    | 3.8    |        |
| U-BOIN                | 4.7    | 6      | 9.8    | 6.5    | 3.2    |        |
| EffTox                | 7      | 6.5    | 21.6   | 9.1    | 7.8    |        |
| AIDE one-stage (U2)   | 3.4    | 4.2    | 8.4    | 8.4    | 5.3    |        |
| AIDE one-stage (U3)   | 3.6    | 4.4    | 8.7    | 8.4    | 4.7    |        |
| AIDE two-stage (U2)   | 3.8    | 4.2    | 7.4    | 7.5    | 6.9    |        |
| AIDE two-stage (U3)   | 3.8    | 4.2    | 7.4    | 7.6    | 6.8    |        |



# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 33           |        |        |        |        |        |        |
| DLT rate              | 0.01   | 0.02   | 0.09   | 0.32   | 0.38   |        |
| Efficacy rate         | 0.05   | 0.08   | 0.2    | 0.4    | 0.5    |        |
| Utility 2             | 0      | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3             | 42.6   | 44     | 48.4   | 51.2   | 54.8   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 12.1   | 10.8   | 30.2   | 30.8   | 10.8   | 5.3    |
| U-BOIN                | 5      | 8.6    | 24     | 32.6   | 29.7   | 0.1    |
| EffTox                | 0      | 0      | 36     | 26     | 26     | 11     |
| AIDE one-stage (U2)   | 9.2    | 5.8    | 11.5   | 42.6   | 30.4   | 0.5    |
| AIDE one-stage (U3)   | 12.7   | 7.5    | 11     | 42.7   | 25.6   | 0.5    |
| AIDE two-stage (U2)   | 8.8    | 4.6    | 12.8   | 39.7   | 33     | 1.1    |
| AIDE two-stage (U3)   | 11.9   | 7      | 13.5   | 40.5   | 25.9   | 1.2    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 4.2    | 5.2    | 7.8    | 8.2    | 4.3    |        |
| U-BOIN                | 4      | 5      | 8.6    | 8.1    | 4.4    |        |
| EffTox                | 4.5    | 3.7    | 17.5   | 13     | 16.6   |        |
| AIDE one-stage (U2)   | 3.3    | 3.6    | 5.9    | 10.3   | 6.8    |        |
| AIDE one-stage (U3)   | 3.3    | 3.7    | 5.9    | 10.5   | 6.3    |        |
| AIDE two-stage (U2)   | 3.4    | 3.6    | 5.1    | 8.2    | 9.6    |        |
| AIDE two-stage (U3)   | 3.4    | 3.6    | 5      | 8.2    | 9.6    |        |
| Scenario 34           |        |        |        |        |        |        |
| DLT rate              | 0.05   | 0.13   | 0.3    | 0.36   | 0.42   |        |
| Efficacy rate         | 0.05   | 0.1    | 0.3    | 0.32   | 0.35   |        |
| Utility 2             | 0      | -0     | 0      | -0     | -0.1   |        |
| Utility 3             | 41     | 40.8   | 46     | 44.8   | 44.2   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 14.7   | 17.8   | 39.1   | 11     | 3.1    | 14.3   |
| U-BOIN                | 6.2    | 12.8   | 29.5   | 27.5   | 23.9   | 0.1    |
| EffTox                | 1      | 7      | 39     | 12     | 11     | 31     |
| AIDE one-stage (U2)   | 12.4   | 8.8    | 40.7   | 23.3   | 5.1    | 9.7    |
| AIDE one-stage (U3)   | 16.2   | 11.2   | 39.9   | 19.7   | 3.1    | 9.9    |
| AIDE two-stage (U2)   | 11.1   | 9.7    | 42.9   | 18.7   | 5.7    | 11.9   |
| AIDE two-stage (U3)   | 15.7   | 12.7   | 38.8   | 17.4   | 3.9    | 11.5   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 5.5    | 7.1    | 8.8    | 4.9    | 2.3    |        |
| U-BOIN                | 5.6    | 7.7    | 8      | 5.4    | 3.5    |        |
| EffTox                | 5.3    | 6.6    | 18.9   | 7.2    | 8.6    |        |
| AIDE one-stage (U2)   | 4.4    | 5.6    | 9.2    | 6.3    | 2.7    |        |
| AIDE one-stage (U3)   | 4.6    | 5.8    | 9.3    | 6.1    | 2.4    |        |
| AIDE two-stage (U2)   | 4.5    | 5      | 9.8    | 5.4    | 3.6    |        |
| AIDE two-stage (U3)   | 4.6    | 5      | 9.8    | 5.5    | 3.5    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 35           |        |        |        |        |        |        |
| DLT rate              | 0.03   | 0.09   | 0.3    | 0.4    | 0.45   |        |
| Efficacy rate         | 0.2    | 0.45   | 0.45   | 0.45   | 0.45   |        |
| Utility 2             | 0.2    | 0.4    | 0.1    | 0.1    | 0      |        |
| Utility 3             | 50.8   | 63.4   | 55     | 51     | 49     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 11.6   | 63.5   | 17.9   | 5.4    | 1.4    | 0.2    |
| U-BOIN                | 10.6   | 61.1   | 19.9   | 6.3    | 2.1    | 0      |
| EffTox                | 41     | 39     | 13     | 4      | 2      | 2      |
| AIDE one-stage (U2)   | 9      | 39.5   | 31.8   | 15.7   | 3.8    | 0.2    |
| AIDE one-stage (U3)   | 12.1   | 46.7   | 27.1   | 10.7   | 3.1    | 0.3    |
| AIDE two-stage (U2)   | 10.1   | 39.5   | 32.2   | 13.5   | 4      | 0.7    |
| AIDE two-stage (U3)   | 15.9   | 43     | 28.3   | 9.4    | 2.7    | 0.7    |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 5.2    | 13.5   | 7      | 3.2    | 1      |        |
| U-BOIN                | 5.7    | 13.1   | 7.7    | 2.9    | 0.7    |        |
| EffTox                | 29.4   | 16.3   | 8.9    | 2.8    | 1.9    |        |
| AIDE one-stage (U2)   | 4.3    | 11     | 8.4    | 4.6    | 1.6    |        |
| AIDE one-stage (U3)   | 4.6    | 12     | 8.1    | 3.9    | 1.3    |        |
| AIDE two-stage (U2)   | 4.5    | 8.2    | 8.9    | 5.1    | 3.2    |        |
| AIDE two-stage (U3)   | 4.5    | 8.2    | 8.9    | 5      | 3.2    |        |
| Scenario 36           |        |        |        |        |        |        |
| DLT rate              | 0.13   | 0.29   | 0.35   | 0.4    | 0.43   |        |
| Efficacy rate         | 0.05   | 0.2    | 0.22   | 0.3    | 0.35   |        |
| Utility 2             | -0.1   | -0.1   | -0.1   | -0.1   | -0.1   |        |
| Utility 3             | 37.8   | 40.4   | 39.2   | 42     | 43.8   |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 12.9   | 33     | 14.3   | 6.4    | 2.2    | 31.2   |
| U-BOIN                | 5      | 19.9   | 17.5   | 23.3   | 33.5   | 0.8    |
| EffTox                | 2      | 9      | 16     | 12     | 11     | 50     |
| AIDE one-stage (U2)   | 9      | 30     | 13.8   | 7.6    | 2      | 37.6   |
| AIDE one-stage (U3)   | 12.3   | 30     | 12.1   | 6.9    | 1      | 37.7   |
| AIDE two-stage (U2)   | 10.5   | 27     | 14.7   | 6.8    | 1.9    | 39.1   |
| AIDE two-stage (U3)   | 14.1   | 28.1   | 11.7   | 5.7    | 1.1    | 39.3   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 6.4    | 8.7    | 5.3    | 3.7    | 1.9    |        |
| U-BOIN                | 7.8    | 7.4    | 5.7    | 4.9    | 4.1    |        |
| EffTox                | 7      | 7.6    | 10     | 5.9    | 8.7    |        |
| AIDE one-stage (U2)   | 5.9    | 8.3    | 5.2    | 3.5    | 1.2    |        |
| AIDE one-stage (U3)   | 6.1    | 8.5    | 5.1    | 3.3    | 1.1    |        |
| AIDE two-stage (U2)   | 6.2    | 7.2    | 5.6    | 2.8    | 1.4    |        |
| AIDE two-stage (U3)   | 6.2    | 7.2    | 5.6    | 2.8    | 1.4    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle\_Max = 2, efficacy futility cutoff = 0.75

|                       | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|-----------------------|--------|--------|--------|--------|--------|--------|
| Scenario 37           |        |        |        |        |        |        |
| DLT rate              | 0.08   | 0.3    | 0.4    | 0.45   | 0.5    |        |
| Efficacy rate         | 0.05   | 0.2    | 0.2    | 0.2    | 0.2    |        |
| Utility 2             | -0     | -0.1   | -0.2   | -0.2   | -0.3   |        |
| Utility 3             | 39.8   | 40     | 36     | 34     | 32     |        |
| OBD selection (%)     |        |        |        |        |        |        |
| BOIN12                | 16.7   | 38.3   | 9.4    | 3      | 0.5    | 32.1   |
| U-BOIN                | 11.4   | 30.2   | 19.4   | 18.6   | 18.7   | 1.7    |
| EffTox                | 1      | 9      | 11     | 3      | 3      | 73     |
| AIDE one-stage (U2)   | 10.2   | 33.7   | 14.3   | 4.1    | 0.5    | 37.2   |
| AIDE one-stage (U3)   | 11.6   | 34.3   | 11.4   | 3.7    | 0.5    | 38.5   |
| AIDE two-stage (U2)   | 14.3   | 32.9   | 11.6   | 3      | 0.5    | 37.7   |
| AIDE two-stage (U3)   | 19.4   | 32     | 8.9    | 1.9    | 0.3    | 37.5   |
| Mean patients treated |        |        |        |        |        |        |
| BOIN12                | 6.9    | 10.4   | 5.2    | 2.6    | 1.2    |        |
| U-BOIN                | 8.1    | 8.4    | 5.6    | 4.3    | 3.6    |        |
| EffTox                | 7.2    | 8.4    | 8.5    | 3.4    | 2.9    |        |
| AIDE one-stage (U2)   | 6      | 9.3    | 5.4    | 3      | 0.9    |        |
| AIDE one-stage (U3)   | 6.1    | 9.5    | 5.2    | 2.8    | 0.8    |        |
| AIDE two-stage (U2)   | 6      | 9.2    | 6      | 2.4    | 0.9    |        |
| AIDE two-stage (U3)   | 6      | 9.2    | 6.1    | 2.4    | 0.9    |        |